OVM Homework Exercises I (WI3405TU)

Submission deadline: 23th October 2026 (electronically via Brightspace)

Please include all answers clearly in your **PDF report** (LaTeX recommended) and attach your code in the appendix. Annotate your code such that one can easily see how your solutions are obtained. Also submit the code separately as a .ipynb, .py, or .m file for implementation checking. Unclear solutions or problems with your code may lead to loss of points. Note that ZIP files will not be accepted (for security and grading reasons). Each question is worth 2 points when done completely correct. When small mistakes are present you can get 1 point and when too many problems are in the solutions no points are given.

1 (2P) Market stock prices:

a. Using the daily close prices (1 year - 1 September 2025 to 31 August 2026) [1])

- Plot the price movement over time.
- Compute the daily return $r_i = \frac{S_i - S_{i-1}}{S_{i-1}}$ and daily logarithmic return $\hat{r}_i = \ln\left(\frac{S_i}{S_{i-1}}\right)$, and create plots to illustrate the patterns of both returns.
- Explain under what circumstances these two types of returns can be considered approximately identical.

b. Statistics:

- Plot the histogram of daily logarithmic returns, compute the sample mean $\hat{\mu}$ and variance $\hat{\sigma}^2$, and produce the quantile-quantile plot, i.e., daily logarithmic returns vs. normal distribution $\mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$.
- Repeat the above process for the weekly close prices (4 years - 2022,2023,2024,2025), and compare the Q–Q plot patterns of weekly and daily returns.

2 (2P). Monte Carlo

Compute the expected value of $1/\sqrt{U}$ for $U \sim U(0,1)$ using Monte Carlo method.

Study the speed of convergence of MC method and whether the errors are asymptotically normal. Discuss your results.

To support your discussion produce figures with estimates and confidence intervals of these estimates for different number of samples $10^2, 10^3, 10^4, 10^5$. To test normality of errors show figures of the histogram together with the normal density with estimated parameters as well as Q-Q plot.

Hint: What can you say about variance of $1/\sqrt{U}$? How should this affect the results?

[1] The data of Apple stock prices can be downloaded from the website `https://www.marketwatch.com/investing/stock/aapl/` under the "Historical quotes" tab.

3 (2P). Simulation of stock price paths.

a. We have the discrete price model $S(t_{i+1}) = S(t_i)(1+\mu\delta t+\sigma\sqrt{\delta t}Z)$, where $Z \sim \mathcal{N}(0,1)$, $T = 1$, $S_0 = 170$, $\sigma = 0.344$, and $\mu = 0.1$.

- Choose the number of time intervals $L = 5$ and plot the price evolution of S_t for $t = i \cdot \Delta t$ with $i = 0, \ldots, L$ and $\Delta t = \frac{T}{L}$.
- Repeat the simulation $M = 5000$ times, and store the simulated S_T to make the histogram.
- Repeat the simulation (above steps) for increasing $L = 10, 20, 50, 100$ to see how the histogram of S_T changes, compared with the theoretical distribution given by the continuous-time stock price model, i.e., $S_T = S_0 e^{(\mu-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z}$. Show the results obtained for different values of L in the form of empirical distribution functions plotted on one figure together the theoretical distribution function.

b. Using the continuous-time stock price model as in a) and $M = 10^2, 10^3, 10^4, 10^5$ simulations, estimate the expectation of S_T with standard Monte Carlo simulation and with the antithetic variables technique. Compute the 95% confidence intervals of your estimates and the ratio their widths I_M/I_M^A.

4(2P). Historical volatility

a. Use the same data as in 1) and compute estimates of historical volatility for the daily and weekly Apple data.

b. Produce figures of daily and weekly volatility estimates using exponentially weighted moving average method (Similar to Figure 20.1 in the book). Use 20 first observations to get initial σ^* and $w = 0.94$.

c. In view of results obtained in b) do you think that the asset pricing model presented during the third lecture is appropriate for this data.

5(2P). Binomial method
In the asset price model, we have the stock price $S_0 = 1.0$, the expiry time $T = 3.0$, $\mu = r = 0.05$, $\sigma = 0.3$, strike price $E = 1.5$. Use $M = 12$ (hence $\delta t = 3/12$) for the valuations.

a. Compute the time-zero price of fixed strike lookback put option with payoff at expiry $\max(E - S^{min}, 0)$ where $S^{min} = \min_{[0,T]}(S_t)$.

b. Determine the time-zero price of a shout put option with payoff

$$\text{Payoff} = \begin{cases} \max(E - S(T), E - S(\tau)), & \text{if holder shouted at time } \tau \text{ (random).} \\ \max(E - S(T), 0), & \text{otherwise.} \end{cases}$$

which allows the holder to shout at most once to the writer before expiration and lock in a payoff of at least $E - S(\tau)$, when $S(\tau) < E$ and $0 \le \tau < T$.